

Technical Project Report: Phishing Email Header & Artifact Analysis Lab

1. Executive Summary & Lab Scope

This technical report details the step-by-step procedure for performing end-to-end phishing email triage and artifact analysis from the perspective of a Tier-1 SOC Analyst.

The primary goal of this lab is to safely acquire, defang, analyze, and document malicious indicators of compromise (IoCs) extracted from suspicious email message files (.eml/.msg), establishing standardized incident response playbooks for enterprise threat containment.

Environment Architecture & Tooling

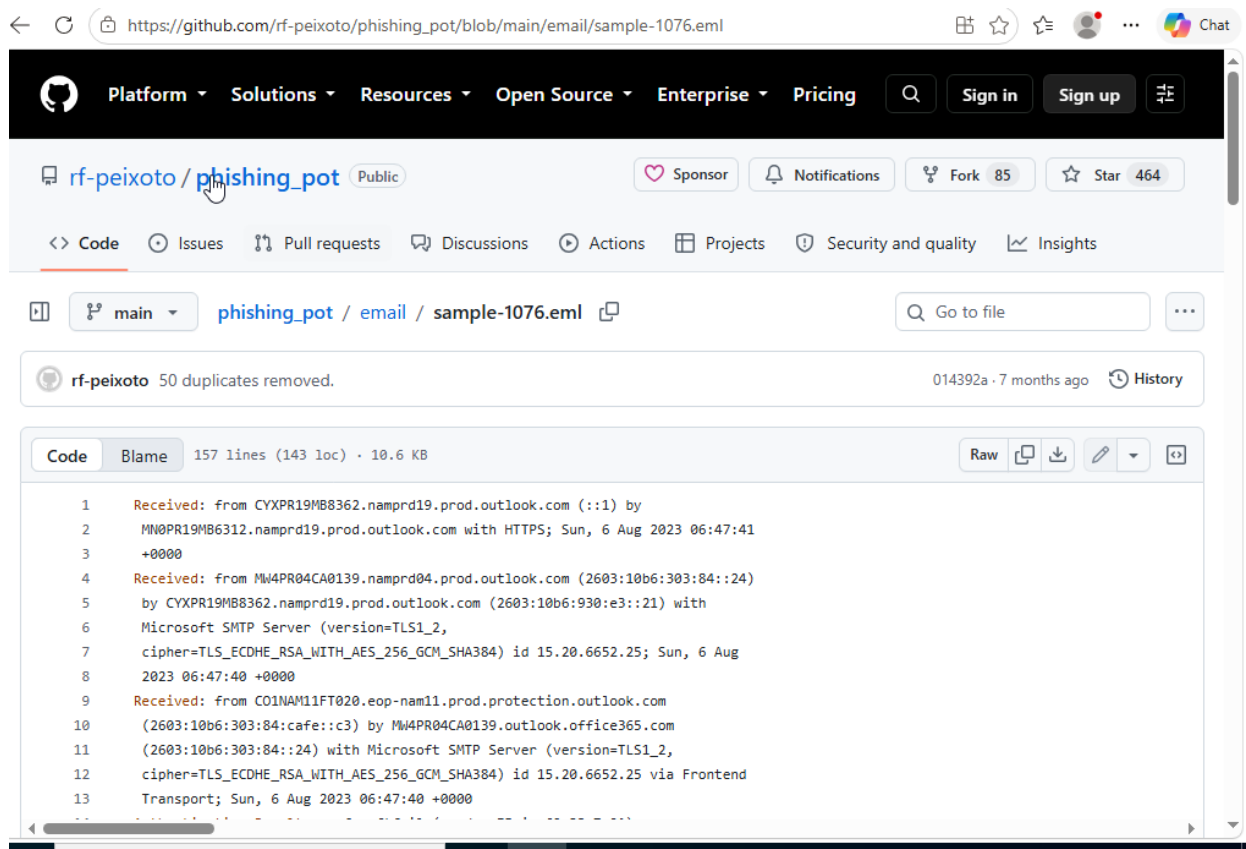
- **Analyst Workstation:** Host OS / Virtual Machine (Isolated Environment)
- **Header Parsing Engines:** Google Admin Toolbox Messageheader / MXToolbox Email Header Analyzer
- **Reputation & Sandbox Analysis:** VirusTotal, URLScan.io, AbuseIPDB
- **Safe Artifact Inspection:** Visual Studio Code / Notepad++ (Plaintext view)
- Raw File Name: sample-1076.eml

2. Execution Phases & Analysis Procedures

Step 1: Obtain a Phishing Sample Safely

To prevent accidental code execution or tracking pixel activation, handle raw email samples within a strictly controlled workflow:

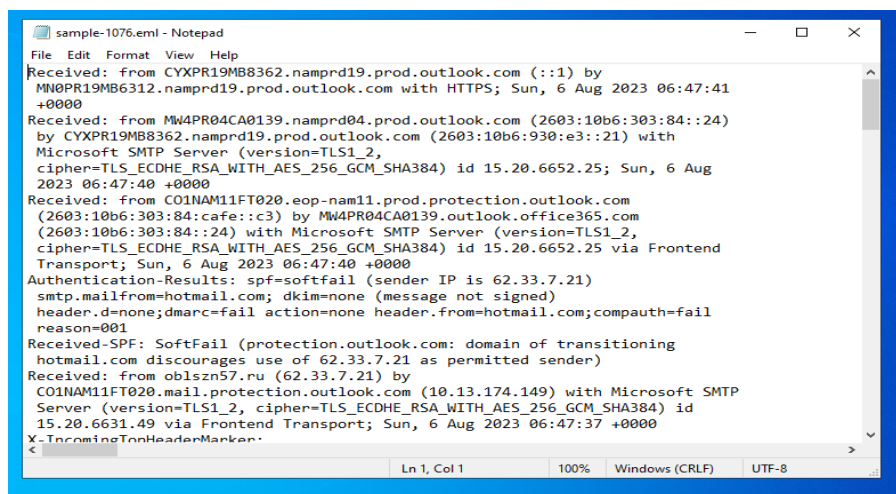
1. Navigate to a public, open-source phishing dataset repository on GitHub (e.g., rf-peixoto/phishing_pot or search for phishing-samples).
2. Download a raw sample saved as a .eml or .msg file.



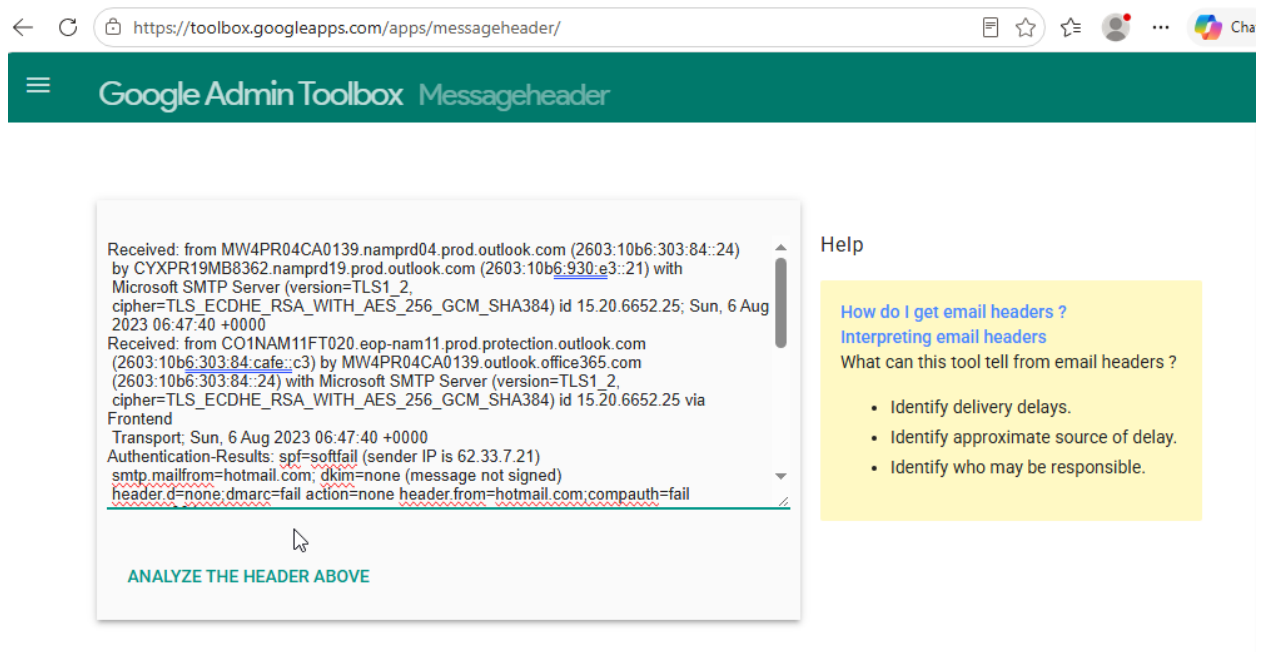
3. **Safety Control:** Do **not** double-click or open the .eml file using an active email client (such as Microsoft Outlook or Apple Mail). Always inspect raw samples using a basic plaintext editor like Notepad or VS Code.

Step 2: Inspect Raw Email Headers

1. Open the downloaded .eml file in your text editor.



2. Locate the header block at the top (everything above the body content, starting with lines like Received:, From:, To:).
3. Copy the entire header text block.
4. Navigate to **Google Admin Toolbox Messageheader** (or **MXToolbox**) and paste the copied header.



5. Analyze the parsed output and extract these mandatory telemetry fields:
 - **Return-Path:** Compare against the From: address to identify domain spoofing discrepancies.
 - **Originating IP Address:** Locate the earliest Received: from entry in the hop chain.
 - **Authentication Status:** Verify pass/fail states for **SPF**, **DKIM**, and **DMARC**.

MessageId	9a424f92-df7b-4732-bf5f-6b223a328066@CO1NAM11FT020.eop-nam11.prod.protection.outlook.com
Created at:	8/1/2023, 5:30:53 PM PDT (Delivered after 4 days)
From:	"Royal Bank Of Canada"<RoyalBanlOfCannada@hotmail.com> Using Microsoft Outlook Express 6.00.2600.0000
To:	Undisclosed recipients,;
Subject:	Re: \$ 12,000,000.00 USD Payment from Royal Bank of Canada
SPF:	softfail with IP Unknown! Learn more
DKIM:	none Learn more
DMARC:	fail Learn more

Step 3: Extract and Defang Artifacts (IoCs)

Before submitting links or binaries to threat intelligence portals, defang all indicators to prevent accidental navigation or execution (e.g., convert http to hXXp and replace dots with [.]).

1. **Extract Embedded Links:** Inspect the body text in your text editor and copy all hyperlinked destination URLs.

```
CO1NAM11FT020.eop-nam11.prod.protection.outlook.com>
```

```
rev.innocent-johnson27@outlook.com>
```

```
CYXPR19MB8362.namprd19.prod.outlook.com
```

```
MN0PR19MB6312.namprd19.prod.outlook.com
```

```
MW4PR04CA0139.outlook.office365.com
```

2. **Extract & Hash Attachments:** If the .eml payload contains an attachment (e.g., .pdf, .zip), save it to an isolated folder and calculate its SHA-256 cryptographic hash via terminal:

- **PowerShell (Windows):**

PowerShell

```
Get-FileHash -Algorithm SHA256 .\suspicious_attachment.pdf
```

- **Bash (Linux/Mac):**

Bash

```
sha256sum suspicious_attachment.pdf
```

Step 4: Threat Intelligence & Reputation Lookup

Run all extracted artifacts through open-source threat intelligence lookup services:

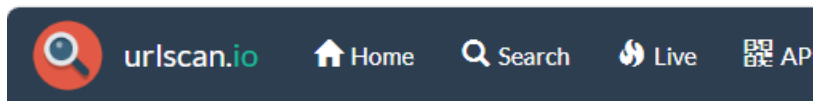
1. **IP Reputation:** Query the originating IP address against **AbuseIPDB** and **VirusTotal** to verify ISP details, geolocation, and reported malicious confidence scores.

The screenshot displays the AbuseIPDB interface for the IP address 62.33.7.21. At the top, a message states "62.33.7.21 was found in our database!". Below this, it indicates "This IP was reported 4 times. Confidence of Abuse is 0%:" followed by a green progress bar at 0%. A table of details follows: ISP is (CN000055) OrelTelecom, Usage Type is Fixed Line ISP, ASN is AS20485, Hostname(s) is oblszn57.ru, Domain Name is orl.ru, Country is Russian Federation (with a flag icon), and City is Orel, Oryol oblast. At the bottom, there is a note about IP info being provided by IPInfo and updated weekly, and two orange buttons labeled "REPORT 62.33.7.21" and "WHOIS 62.33.7.21".

ISP	(CN000055) OrelTelecom,
Usage Type	Fixed Line ISP
ASN	AS20485
Hostname(s)	oblszn57.ru
Domain Name	orl.ru
Country	Russian Federation
City	Orel, Oryol oblast

2. **URL Sandbox Execution:** Submit defanged links to **URLScan.io** and **VirusTotal**.

- *URLScan.io* executes the link inside an isolated sandbox, capturing DOM structures, HTTP request trees, and landing page screenshots without risk to your local machine.



62.33.7.21

Private Scan



URL: <http://62.33.7.21/>

Submission: On September 23 via manual (September 23rd 2026, 2:42:55 pm UTC) fr

We could not scan this website!

This can happen for multiple reasons:

- The site could not be contacted (DNS or generic network issues)
- The site uses insecure TLS (weak ciphers e.g.)
- The site requires HTTP authentication

Take a look at the [JSON output](#) or the screenshot to determine a possible cause.





http://62.33.7.21/

Sign inSign up

0
/ 90
Community Score

URL report for 2026-09-23 14:56:26 UTC.

ReanalyzeMore

http://62.33.7.21/	Status	Content Type	Last Analysis Date
62.33.7.21 62.33.7.21	-	-	a moment ago

SUMMARYDETECTIONDETAILSCOMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Table of Contents

- History

History

- First Submission2026-09-23 14:56:26 UTC - UNKNOWN OR UNSPECIFIED
- Last Submission2026-09-23 14:56:26 UTC - UNKNOWN OR UNSPECIFIED
- Last Analysis2026-09-23 14:56:26 UTC

←↻🔒https://www.virustotal.com/gui/ip-address/62.33.7.21/details



URL, IP address, domain or file hash

Sign inSign up

0
/ 89
Community Score

No security vendor flagged this IP address as malicious

ReanalyzeMore

62.33.7.21 (62.33.0.0/16)

RULast Analysis Date29 days ago

AS 20485 (Joint Stock Company TransTeleCom)

DETECTIONDETAILSRELATIONSCOMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Basic Properties

- Network62.33.0.0/16
- Autonomous System Num...20485
- Autonomous System Label...Joint Stock Company TransTeleCom
- Regional Internet RegistryRIPE NCC
- CountryRU
- ContinentEU

Last HTTPS Certificate

3. **Payload Hash Verification:** Query the calculated SHA-256 hash in **VirusTotal** to check multi-vendor antivirus detection ratios.

SHA-256

```
PS C:\Users\Administrator\Downloads> Get-FileHash "sample-1076.eml" -Algorithm SHA256
```

Algorithm	Hash	Path
SHA256	B632BA99FA67C19C3EC0F1A803BD9E2C2B424992AE00B89379732F212226814F	C:\Users\Administrator\Downlo...

```
PS C:\Users\Administrator\Downloads> _
```

The screenshot shows the VirusTotal web interface for the file `sample-1076.eml`. The file's SHA-256 hash is `b632ba99fa67c19c3ec0f1a803bd9e2c2b424992ae00b89379732f212226814f`. The interface displays a green circle with the number 0, indicating no security vendors flagged the file as malicious. The file size is 10.63 KB, and the last analysis date was 1 year ago. The file type is identified as an email. The interface also shows a community score of 0/63 and a link to join the community.

DETECTION	DETAILS	RELATIONS	BEHAVIOR	COMMUNITY
Basic properties				
MD5	dbe16107f9968398df043a410b991a67			
SHA-1	b2ef4cd5393ea6969c981594c5d6d794d53cb1bb			
SHA-256	b632ba99fa67c19c3ec0f1a803bd9e2c2b424992ae00b89379732f212226814f			
SSDEEP	192:A643lrmvQNvCirIjuAbAyQsENY7iyUOdKpCudQEceTX5QXVe00Q6KBvMZYa:K3lrmvctksyQsENY7iyUNPc6Dcl5QXAV			
TLSH	T191221B95524A31359EF202817BC9BE42E76165A87BB2A0E4B6F4D010174F563EF2B1CF			
File type	Email (internet, email)			
Magic	RFC 822 mail, ASCII text, with very long lines (960u), with CRLF line terminators			
TrID	E-Mail message (Var. 5) (100%)			

3. Incident Triage & Response Documentation

Incident Report Template

Header Information

- **Incident ID:** INC-20260923-01
- **Severity:** **HIGH** (*Financial impersonation/Advance-fee fraud targeting users*)
- **Analyst Name:** Oluwatoni Oderinlo

- **Date:** September 23, 2026

1. Incident Overview

- **Subject Line:** Re: \$ 12,000,000.00 USD Payment from Royal Bank of Canada
- **Claimed Sender:** "Royal Bank Of Canada" RoyalBanOfCannada@hotmail.com
- **Actual Sender:** oblszn57.ru (via Postfix mail server)

2. Authentication Results

- **SPF: SoftFail** (outlook.com: domain of transitioning hotmail.com discourages use of 62.33.7.21 as permitted sender)
- **DKIM: None** (Message not digitally signed)
- **DMARC: Fail** (action=none header.from=hotmail.com; compauth=fail reason=001)

3. Indicators of Compromise (IoCs)

- **Sending IP:** 62.33.7.21 (AS 20485 - Joint Stock Company TransTeleCom, Russian Federation)
- **Malicious URL:** None found in the raw text body; email relies strictly on direct reply-to social engineering manipulation (rev.innocent-johnson27@outlook.com).
- **Payload Hash (SHA-256):**
b632ba99fa67c19c3ec0f1a803bd9e2c2b424992ae00b89379732f212226814f

4. Analysis Summary

The analyzed email (sample-1076.eml) is a high-severity advance-fee/financial impersonation scam spoofing the "Royal Bank of Canada". Header analysis confirms domain spoofing: the actual originating server is oblszn57.ru running Postfix, and the sending IP (62.33.7.21) originates from TransTeleCom in the Russian Federation. Email authentication protocols failed, showing an SPF SoftFail, missing DKIM signature, and a DMARC Fail. While no malicious links are embedded in the body, the threat relies on social engineering to trick victims into replying directly to rev.innocent-johnson27@outlook.com. The raw payload hash was identified as b632ba99fa67c19c3ec0f1a803bd9e2c2b424992ae00b89379732f212226814f.

5. Recommended Remediation & Playbook Steps

1. **Network Layer:** Block originating IP 62.33.7.21 at the perimeter firewall/mail gateway.

2. **Email Gateway Layer:** Add oblszn57.ru and the address rev.innocent-johnson27@outlook.com to the organization's email transport blocklist.
3. **Tenant Containment:** Purge all instances of emails matching the subject line Re: \$ 12,000,000.00 USD Payment from Royal Bank of Canada across all enterprise mailboxes.
4. **User & Endpoint Remediation:** Identify any users who replied to the suspicious message, isolate their endpoints, force a credential reset, and flag their accounts for elevated monitoring.

Comprehensive Troubleshooting Framework

- **Issue A: Missing Headers During Plaintext Inspection**
 - *Cause:* Email clients or default text editors auto-formatting or stripping raw ASCII/header data.
 - *Resolution:* Ensure .eml files are strictly opened in plaintext editors like Notepad++ or VS Code with UTF-8 encoding. Never open .eml files in full desktop email clients (e.g., Outlook) to avoid automatic loading of scripts or remote resources.
- **Issue B: Absence of Embedded Links or Attachments**
 - *Cause:* Attackers utilizing pure social engineering/advance-fee fraud techniques designed to bypass URL sandbox scanners and payload signatures.
 - *Resolution:* Shift focus to header authentication evaluation (SPF, DKIM, DMARC alignment), originating IP intelligence lookups via AbuseIPDB/VirusTotal, and inspecting Reply-To/Return-Path address discrepancies.
- **Issue C: Ambiguous DMARC Pass/Fail Metric**
 - *Cause:* The From: header display address differs from the underlying server connection domain.
 - *Resolution:* Verify alignment between the Header From (hotmail.com) and the actual envelope sender/server host (oblszn57.ru). If domains fail to align despite individual SPF checks, treat DMARC as Fail.

5. Curated Reference Materials & Repositories

Technical Analysis Tools

- [Google Admin Toolbox Messageheader](#) – Native header parsing utility.
- [MXToolbox Email Header Analyzer](#) – Graphical header route diagnostic engine.
- [URLScan.io Submission Portal](#) – Free URL sandbox & web traffic scanner.
- [VirusTotal Intelligence Search](#) – Multi-engine file hash and IP reputation database.

Frameworks & Sample Datasets

- [Phishing Pot Sample Repository \(GitHub\)](#) – Public archive of real-world phishing email .eml files.
- [CISA Technical Guide: Email Authentication Standards](#) – Best practices for implementing SPF, DKIM, and DMARC enforcement.